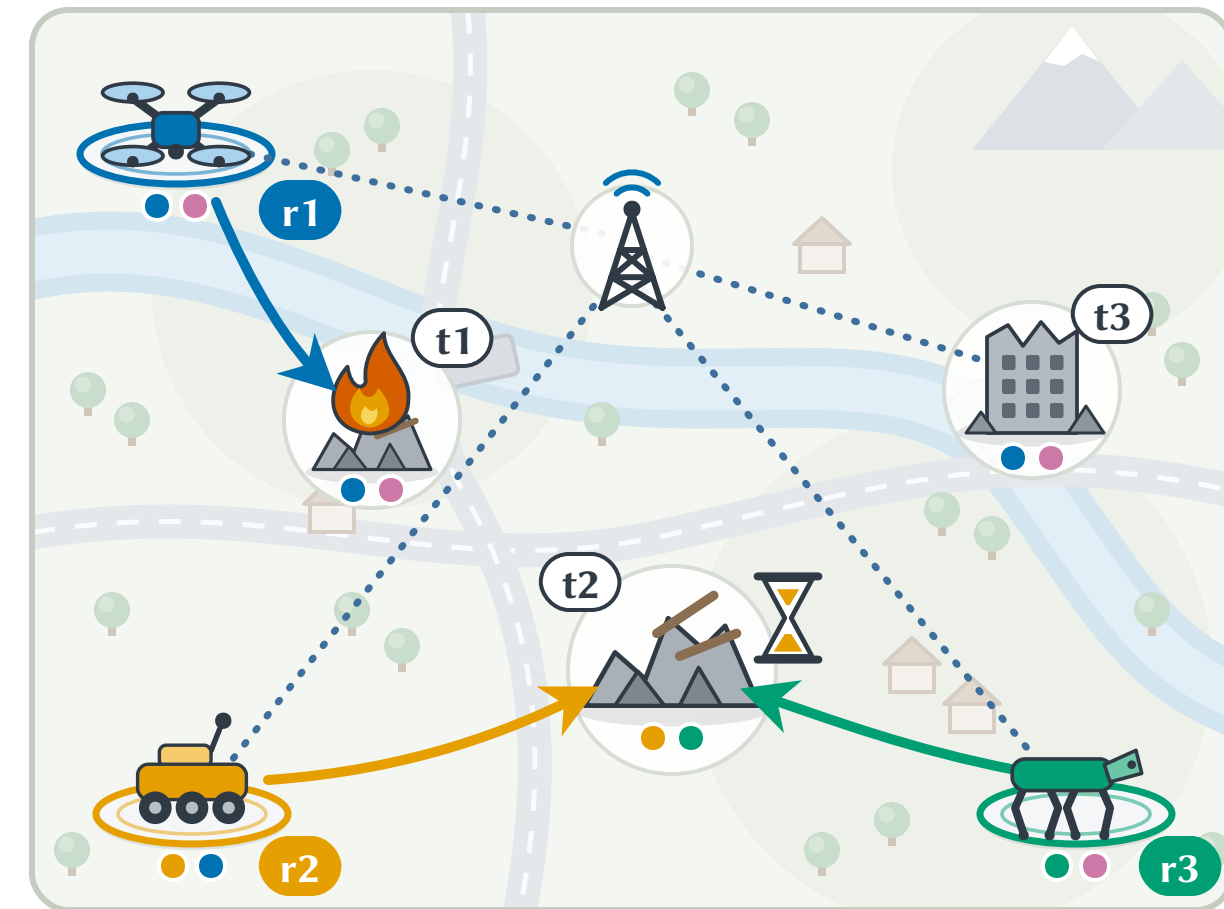
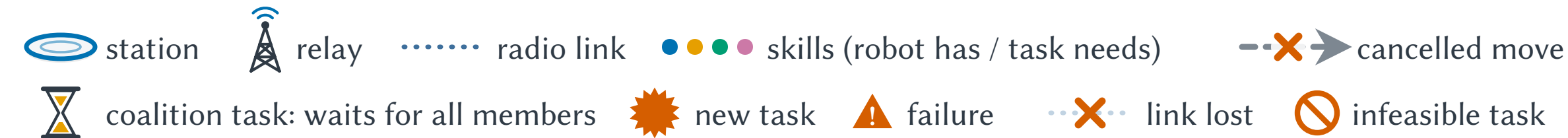
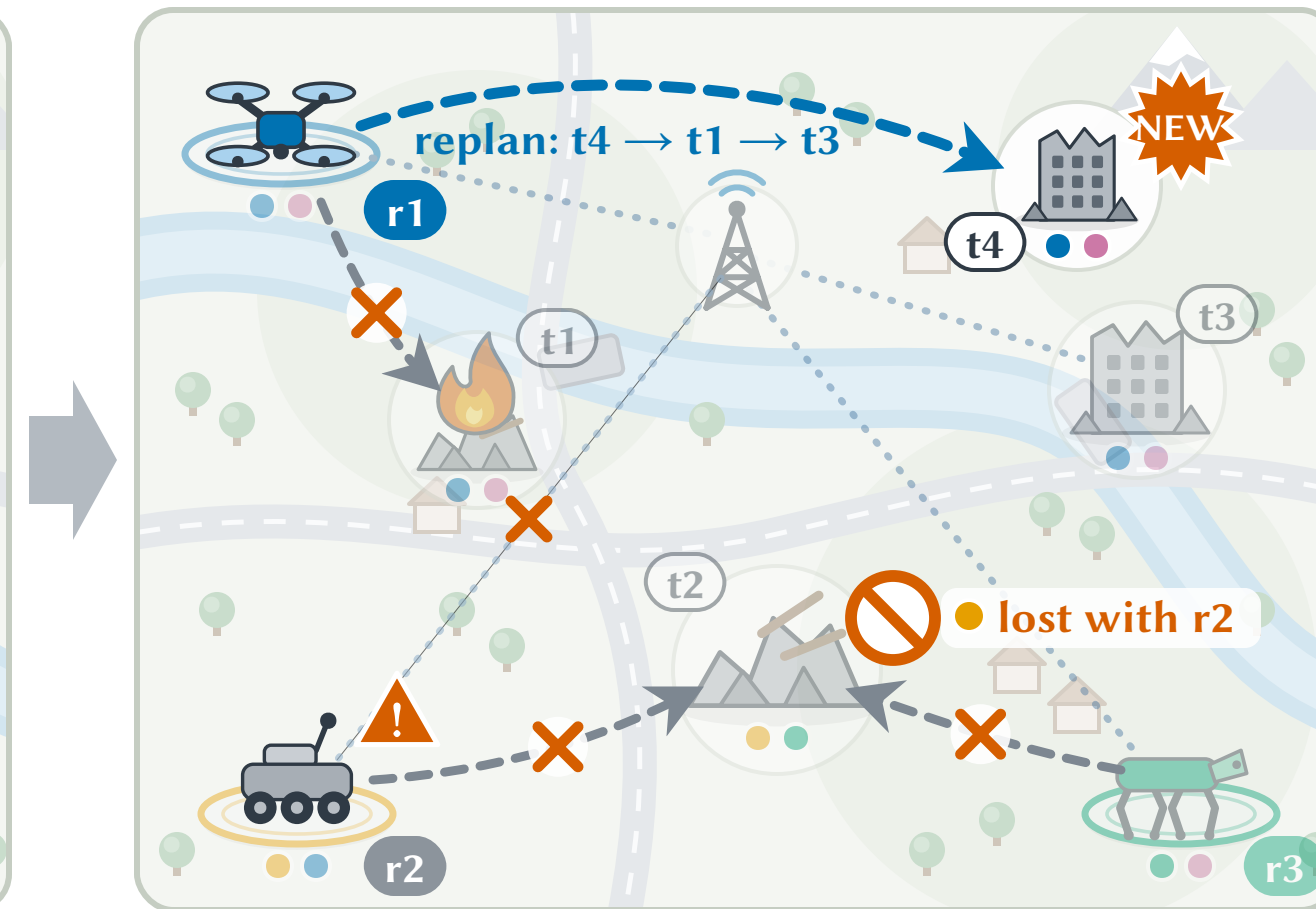


(a) Static vs. dynamic

static: all tasks known at start

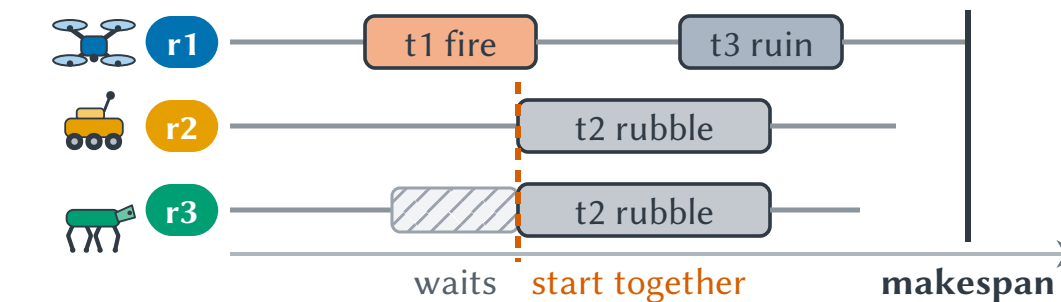
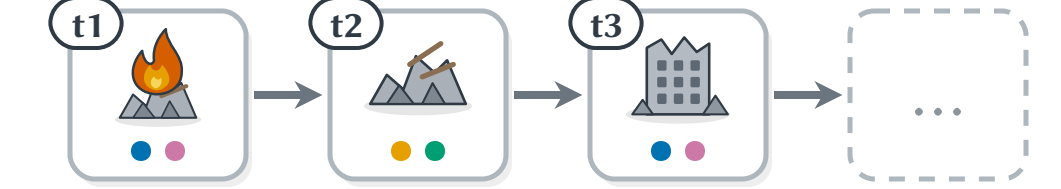


dynamic: tasks appear, robots fail

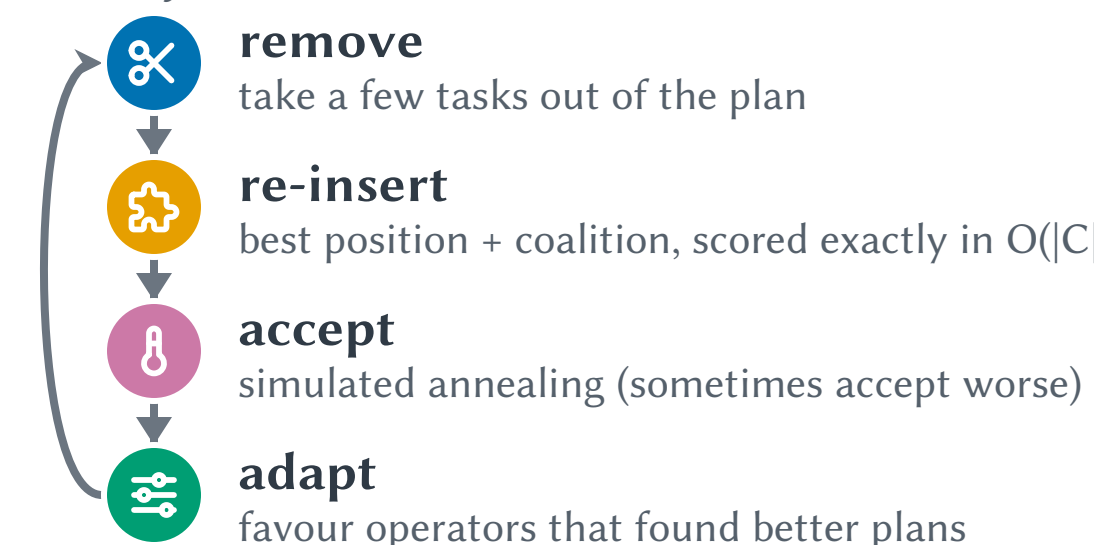


(b) Our search: ALNS

plan = one global task order + a minimal coalition per task



robots follow the order: no deadlock, exact schedule



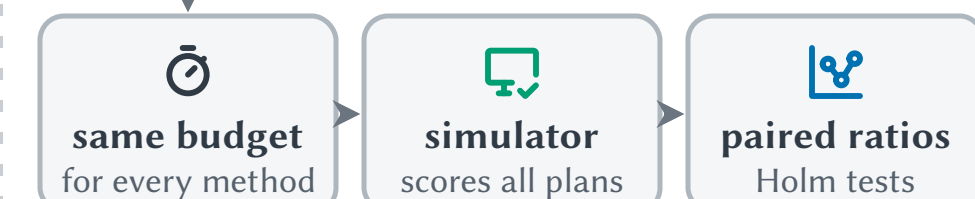
(c) Fair test

protocol

- 1 Tune**
on development set
- 2 Dry run**
on validation set
- Pre-register**
freeze the protocol
- 4 Test**
once, on test set

methods

- ALNS (ours)**
search on coalitions
- RL policy**
published, sampled
- CP-SAT**
full model, 2 LNS
- MILP**
CTAS-D, exact
- Greedy restarts**
8 randomized runs



test set

8 settings	400 instances	25–150 robots	50–500 tasks
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